

Data Science

CSE558

Recap

- Aposteriori Estimation

Overview

- Concentration Inequalities

Exercise

You have signed a contract with a company, who will provide you captcha for your website. The company claims to have a database of 1M images and for every security check it randomly selects one from it.

How can you verify the claim of its database size?

Idea: check for duplicate captchas in some security tests.

If there are n unique captchas and you conduct m tests then how many pairwise duplicates over expectations?

Markov's Inequality

Let $\mathbf{X}$ be a random variable that takes non negative values. Then for all $\mathbf{a} > 0$,

$$\mathbf{P}(\mathbf{X} \geq \mathbf{a}) \leq \frac{\mathbf{E}[\mathbf{X}]}{\mathbf{a}}$$

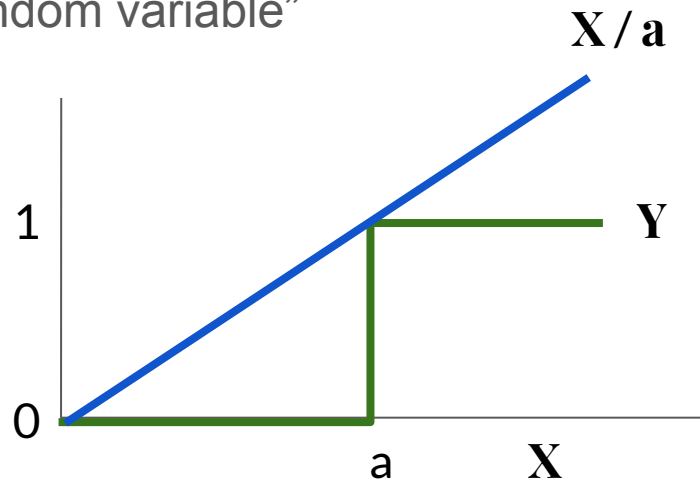
Proof: Consider the following “indicator random variable”

$$\mathbf{Y} = \begin{cases} 1, & \text{if } \mathbf{X} \geq \mathbf{a} \\ 0, & \text{otherwise} \end{cases}$$

So, $\mathbf{E}[\mathbf{Y}] = \mathbf{P}(\mathbf{X} \geq \mathbf{a})$

Notice that $\mathbf{Y} \leq \mathbf{X}/\mathbf{a}$

$$\mathbf{P}(\mathbf{X} \geq \mathbf{a}) = \mathbf{E}[\mathbf{Y}] \leq \mathbf{E}[\mathbf{X}/\mathbf{a}] = \mathbf{E}[\mathbf{X}]/\mathbf{a}$$



Example

A biased coin that lands heads with probability 0.2 is flipped 100 times independently. With what probability it lands heads at most 50 times?

Solution: Consider the random variable, $\mathbf{X} = \begin{cases} 1, & \text{if heads} \\ 0, & \text{if tails} \end{cases}$

Now, $E[\mathbf{X}] = \mathbf{X}(\text{heads}) \cdot P(\text{heads}) + \mathbf{X}(\text{tails}) \cdot P(\text{tails}) = 1 \cdot 0.2 + 0 \cdot 0.8 = 0.2$

So, $100 \cdot E[\mathbf{X}] = 20$.

$P(\text{\#heads} > 50) \leq 100 \cdot E[\mathbf{X}]/50 = 0.4$

With 0.6 probability, the coin lands head at most 50 times.

Example

Given an unbiased die of k faces, what is the probability that you see each of the k faces at least once, in at most $3k \cdot \ln(k)$ random rolls of the die?

Solution: ϵ : The number of rolls you need, to see each of the k faces at least once.

$E[\epsilon] = k \cdot \ln(k)$, using arguments similar to coupon collector problem.

Now using, Markov's inequality we get, $\Pr(\epsilon > 3k \cdot \ln(k)) \leq \frac{1}{3}$.

So, with at least $\frac{2}{3}$ probability you will see each of the k faces at least once in a $3k \cdot \ln(k)$ random rolls of the die.

Example

Back to captcha problem. For 1M claimed captchas if you check with 1000 of them then what is the probability of getting at least 10 duplicates?

Example

Your friend tells you that he had four job interviews last week. He says that based on how the interviews went, he thinks he has a 20% chance of receiving an offer from each of the companies he interviewed with. Nevertheless, since he interviewed with four companies, he is 90% sure that he will receive at least one offer. Is he right?

Markov's Inequality

Is it weak? Give an example, where markov's inequality is weak.

Given an unbiased coin what is the probability that out 20 independent tosses we see at least 16 heads.

Moments of Random Variable

The k^{th} moment of a random variable X is $E[X^k]$

Moments uncovers more informations about the random variable.

First moment is the expectation of the RV.

We get variance of RV using the first and the second moment of RV.

$$\text{Var}[X] = E[(X - E[X])^2] = E[X^2] - (E[X])^2 \text{ (Prove it)}$$

$$\text{Standard Deviation: } \text{Sd}[X] = (\text{Var}[X])^{1/2}$$

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]?$$

Linearity of Variance

For any two random variable **X** and **Y**,

- **$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] + 2 \cdot \text{Cov}(X,Y)$** , where
- **$\text{Cov}(X,Y) = E[(X - E[X]) \cdot (Y - E[Y])]$**

If **X** and **Y** are independent random variable then,

- **$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y]$**

Chebyshev's Inequality

Let $\mathbf{X}$ be a random variable. Then for any $\mathbf{a} > 0$,

$$\mathbf{P} (|\mathbf{X} - \mathbf{E}[\mathbf{X}]| \geq \mathbf{a}) \leq \frac{\mathbf{Var}[\mathbf{X}]}{\mathbf{a}^2}$$

Proof: Recall, that a random variable does not affect the probability distribution.

Consider the non-negative random variable $\mathbf{Y} = (\mathbf{X} - \mathbf{E}[\mathbf{X}])^2$

Apply, Markov's inequality on $\mathbf{Y}$.

$$\mathbf{P} (|\mathbf{X} - \mathbf{E}[\mathbf{X}]| \geq \mathbf{a}) = \mathbf{P} (\mathbf{Y} \geq \mathbf{a}^2) \leq \frac{\mathbf{E}[\mathbf{Y}]}{\mathbf{a}^2} = \frac{\mathbf{Var}[\mathbf{X}]}{\mathbf{a}^2}$$

Example

A biased coin that lands heads with probability 0.2 is flipped 100 times independently. With what probability it lands heads at most 50 times?

Solution: Consider the random variable, $\mathbf{X} = \begin{cases} 1, & \text{if heads} \\ 0, & \text{if tails} \end{cases}$

$$\mathbf{E}[\mathbf{X}^2] = \mathbf{X}(\text{heads})^2 \cdot \mathbf{P}(\text{heads}) + \mathbf{X}(\text{tails})^2 \cdot \mathbf{P}(\text{tails}) = 1^2 \cdot 0.2 + 0^2 \cdot 0.8 = 0.2$$

$$\text{So, } \mathbf{Var}[\mathbf{X}] = \mathbf{E}[\mathbf{X}^2] - (\mathbf{E}[\mathbf{X}])^2 = 0.2 - 0.2^2 = 0.16$$

$$\text{So, } 100 \cdot \mathbf{Var}[\mathbf{X}] = 16.$$

$$\mathbf{P}(\# \text{heads} > 50) = \mathbf{P}(|\# \text{heads} - 20| > 30) \leq 100 \cdot \mathbf{Var}[\mathbf{X}] / 30^2 \approx 0.017$$

With 0.983 probability, the coin lands head at most 50 times.